

Table A3: Rater 2's evaluation of the reviewed studies across five methodological rigour criteria: detection performance, theoretical foundations, computational efficiency, interpretability, and ecological relevance. Each criterion is scored on a scale of 0 (very poor) to 3 (excellent).

Paper	Detection Performance Score	Theoretical Foundation Score	Computational Efficiency Score	Interpretability Score	Ecological Relevance Score
Zaza et al. [30]	2 NN-AE achieves 100% recall (zero false normals) with a negligible 0.35% false anomaly rate; precision, recall, specificity, F1, AUC, and confusion matrices are reported across three resampling conditions with confidence intervals, and a dedicated threshold-optimisation algorithm is validated against these metrics.	2 Grounded in well-established methods (autoencoder reconstruction error theory, Isolation Forest, LOF, DBSCAN), with formal justification, equations, and hyperparameter search (GridSearch) documented for each model.	2 Training time vs. data volume is explicitly benchmarked; NN-AE scales acceptably but requires substantially more compute than IF/LOF/DBSCAN, and the authors flag this as a limitation for resource-constrained settings; not designed for real-time use.	2 Three complementary mechanisms: feature-wise reconstruction error analysis identifying most influential features; SHAP analysis for IF providing game-theoretic feature importance; clear ecological guidelines translating outputs into actionable decisions. The anomaly types (single station, >120 days stationary, missed consecutive stations) are explicitly defined with ecological interpretations.	3 First automated ML/DL approach for telemetry anomaly detection; 50 tagged fish, 16 receivers, 3+ million detections over 5 years; directly improves data integrity for fish movement studies supporting fisheries management; open source code and data on GitHub; practical guidelines provided.
Kareinen et al. [31]	3 Reports 5 metrics for all methods. Best results: CAC achieves Open-Set F-score 95.8% on zooplankton and 93.9% on phytoplankton. Systematic comparison of 5 model variants. 5-fold cross-validation repeated for robustness.	2 Three well-established OSR methods: OpenMax with Weibull fitting for unknown class probability adjustment; ArcFace with additive angular margin loss for metric learning; Class Anchored Clustering with anchored class centers in feature space. Novel contribution: quantile-based class specific threshold selection requiring only one tunable parameter. First application of CAC to plankton recognition.	1 ResNet-18 and DenseNet-121 backbones pre-trained on ImageNet. Acknowledges CSC computational resources. No specific training time, inference speed, or memory footprint reported. Standard deep learning GPU requirements; feasible for batch processing but not characterised for real-time or edge deployment.	1 OpenMax provides partial interpretability through Weibull fitting parameter and MAV distance analysis. Threshold sensitivity analyses show how performance varies across threshold/quantile values. However, no attention maps, Grad-CAM, or feature importance analyses included. ArcFace and CAC embedding spaces remain opaque. Class specific threshold visualisation provides some diagnostic insight.	3 Directly addresses open-set plankton recognition for operational monitoring. Uses two real Baltic Sea datasets from the Finnish Environment Institute. Explicitly designed for in-situ imaging instruments encountering novel species and non-plankton particles. Data made publicly available. Directly supports HAB monitoring and autonomous ocean observation. Acknowledges fine-grained nature of plankton OSR as a key ecological challenge.
Siddagangaiah [40]	2 iForest detected 15 of the 17 labelled anomalies in the low-frequency band with 75% precision, but precision drops to 53.8% and 50% in mid and high-frequency bands (F1: 63.6% and 66.66%). No AUROC reported, ground truth labelling methodology is unclear.	3 iForest has solid foundations in random partition trees; NeuralProphet, NHiTS, and TiDE represent three theoretically distinct deep learning forecasting approaches; Optuna hyperparameter tuning is grounded in Bayesian optimisation.	2 Hyperparameter tuning times are reasonable (9-47 min per model on RTX 4090), but the full multi-stage pipeline requires substantial resources; unsuitable for lightweight or embedded deployment.	2 Three frequency bands are ecologically interpretable; trend decomposition separates annual, seasonal, and diurnal components; correlation analysis compares NeuralProphet trends with singular spectrum analysis; per-band precision/recall enables diagnostic assessment.	3 Directly addresses marine noise pollution, fish and shrimp population monitoring, and climate change effects on soundscapes; data available on Zenodo; supports marine spatial planning and conservation.

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Pu et al. [38]	2 AUROC 96.3% on DYB-PlanktonNet, above OE, CAC and OECC, but leaves some room on TNR95 (79.9%). The paper does not report precision or recall, and lacks thorough diagnostic analysis of failure modes.	2 Synthesises information theory (cross-entropy), variational inference (KL divergence for uncertainty), and metric learning (Anchor loss for embedding separation) into a coherent framework, with each component well-justified.	2 A Wide ResNet (40 layers) is lightweight, and the combined loss adds little overhead compared with OECC, which trains the network twice, but no inference time is reported for real-time in situ plankton monitoring.	2 The Wide ResNet operates as a black-box feature extractor; no attention maps or saliency visualisations provided. KL divergence offers limited uncertainty quantification but does not explain which image regions drive decisions.	3 Targets in situ plankton monitoring, essential for continuous ocean observation. Treating the 41 plankton classes as normal and particles and bubbles as anomalies reflects the reality of coastal samples, where non-plankton particles are reported to exceed 97%.
Beyan [33]	2 Extensive evaluation across three methodological approaches using Wilcoxon signed-rank, Friedman, Holm, and paired t-tests, with true positive rate, true negative rate, geometric mean and AUC reported and used for validation. Evaluation spans fish trajectory datasets and public imbalanced datasets, with active learning integration also assessed. The best parameters are taken on the test folds, so validation is not fully independent.	3 Integrates clustering theory, outlier detection, hierarchical classification, active learning, and feature selection. The 21 rule-based filters encode ethological domain knowledge; the hierarchical decomposition represents a novel theoretical contribution.	2 Rule-based filtering is fast, but SFFS and hierarchical decomposition are computationally demanding. Real-time deployment would require further optimisation.	2 The rule-based approach provides fully transparent decisions; SFFS reveals discriminative features; the hierarchical tree structure enables tracing of classification paths; cluster cardinalities indicate relative unusualness.	3 Explicitly motivated by assisting marine biologists in detecting environmental change indicators; collected the largest labelled fish trajectory dataset at the time; designed with marine ecologists as end users.
Walker and Orenstein [34]	2 F1 score and accuracy are reported for both OOD detection and dataset shift scenarios across multiple imbalance ratios, with standard deviations. However, precision and recall are not reported separately, and the 95% recall threshold calibration on validation data limits independent metric reporting.	2 Grounded in Outlier Exposure theory with a formalised alternating optimisation using KL divergence, but key parameters are set empirically and no formal theoretical justification is provided for why hard negatives specifically improve OOD performance.	2 Fine-tuned ResNet-18 keeps the model lightweight, but the two-phase alternating optimisation adds training overhead. No inference time or scalability metrics are reported for real-time deployment scenarios.	2 Standard deep learning black-box. No attention maps, saliency visualisations, or feature importance analysis. Background image weights provide limited insight into which features drive decisions.	3 Addresses a genuine ecological challenge of tracking rare plankton taxa under realistic open-ocean conditions including dataset shift and novel species, but stops at classification performance with no downstream ecological interpretation or operational deployment demonstrated.
Wang et al. [39]	3 Precision/recall/F1 reported per class, validated across models and ablations; Lite3D achieves 99% F1-score.	2 Grounded in established 3D-CNN theory, with full mathematical derivation of layer sizing and loss function.	3 Highly efficient, Lite3D is 50 times smaller in model size and 57 times lighter in trainable parameters compared to 3DCNN and C3D.	2 PCA visualisation shows feature importance, but the model remains a black box with no per-decision explanation.	3 Detects disease/stress-linked behaviors in real fish, directly supporting aquaculture and rehabilitation monitoring.
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Chen and Zhu [36]	3 ROC-AUC of 0.9843 (vs. 0.9751 for baseline CTC-KD); precision 0.99 (abnormal) and 0.85 (normal); recall 0.96 (abnormal) and 0.95 (normal). Evaluation spans three anomaly types: aberrant crystallization, other species, and abnormal coloration, with an ablation study, although the threshold and epoch were chosen on the test images.	2 Grounded in Hinton et al.'s knowledge distillation framework with a clear theoretical innovation, asymmetric inputs (color teacher, grayscale student) force greater embedding discrepancy for anomalies. The dual-loss formulation (Lval Euclidean distance + Ldir direction cosine) is theoretically coherent.	2 Training runs on a single NVIDIA RTX2060 GPU with batch size 6 and images resized to 128×128 px, which is modest and feasible. However, no inference time, FPS, or deployment cost is reported, and scalability to large otolith databases is not evaluated.	2 Gradient-based anomaly heatmaps are generated and shown per anomaly type, giving spatial localisation of detected anomalies. However, the explanations are pixel-level only; no ecological interpretation of why or how specific features drive anomaly scores is provided.	3 Otolith analysis shows fish age, growth, population dynamics, and stock assessment; anomalous otoliths introduce systematic bias; first otolith anomaly detection dataset proposed; flagging problematic specimens before analysis has immediate practical implications for fisheries management.
Beyan et al. [32]	2 Outlier trajectory removal validated against 1000 manually examined trajectories; catches only 52% of erroneous trajectories and removes 22% of correct ones, with no precision or F1 reported.	3 Strong grounding: mcRBM, cubic B-spline control-point trajectory representation with explicit justification over HMM/PCA/vector quantisation, and outlier-via-cluster-cardinality adapted from prior work. Each design choice is argued.	2 Key selling point is that mcRBM needs no pairwise distance calculations ($O(N)$ parameters rather than $O(2^N)$), so it scales to 3.6M trajectories where k-means/hierarchical wouldn't. But training runs 900 to 1600 epochs, it's offline only, and no runtime/scalability numbers are given; acceptable and scalable, not real-time.	2 Per-cluster control-point distributions are shown with distinctive patterns and low intra-cluster variance, giving some insight. But the mcRBM latent activations that define clusters are fairly opaque; you see which trajectories are outliers more than why.	3 Directly targets fish speed vs. water temperature as a climate-sensitivity signal for coral-reef fish, on real Fish4Knowledge video, complementary to marine-biology findings. Clear ecological application.
Ward et al. [37]	1 Detects temporal and spatial anomalies using Bayesian spatiotemporal models and ARIMA. However, no standard AD metrics (e.g., precision, recall, F1-score) are reported.	2 Uses well-established Bayesian spatiotemporal GLMMs, Gaussian processes, MCMC, and ARIMA with strong theoretical support.	1 Suitable for large ecological datasets but computationally intensive due to Bayesian MCMC estimation and not designed for real-time analysis.	2 Anomalies are clearly linked to environmental variables and spatial patterns, making the results easy to interpret ecologically. But interpretability is limited by model complexity.	3 Directly applies anomaly detection to assess fish population responses to the Deepwater Horizon oil spill, providing high ecological relevance.
Bilik et al. [29]	3 The paper reports comprehensive quantitative metrics including F1 score, precision, recall, and AUC for all nine phytoplankton species. The best overall combination achieves an F1 score of 0.75 across all classes, with species-specific F1 ranging from 0.60 (Centrales) to 0.83 (Peridiniella Single).	2 The VQVAE architecture is well-grounded in generative modelling literature, and the paper provides ecological justification for the anomaly detection formulation; parasites deviate from healthy phytoplankton training data, causing poor reconstruction. However, the theoretical contribution is methodological rather than foundational.	2 The paper systematically evaluates computational trade-offs across architectures but does not report specific training times, inference speeds, or memory footprints. The TensorFlow 2 implementation is publicly available.	2 The reconstruction-difference approach provides intuitive visual interpretability; the 2D feature space visualisation shows OK/NOK cluster separation; however, no systematic feature importance analysis, attention maps, or saliency methods are included.	3 The paper addresses a genuine practical need where manual inspection of imaging flow cytometry data is infeasible at scale.

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Shutler et al. [35]	2 68.0% correct classification and false alarm rate of 0.24 are reported with a binary confusion matrix, but validation rests on only n=25 matchups, standard metrics like F1-score are absent, and cloud cover and temporal mismatches limit evaluation thoroughness.	2 Built on established statistical background subtraction and temporal variance modelling from Shutler et al. (2010) and Gohin et al. (2008), but several key parameters are carried over empirically from prior work without re-derivation or sensitivity testing for the HAB context.	2 Processes per-pixel statistics over a 55-day rolling window on large-scale satellite imagery, suitable for broad monitoring, but no runtime or scalability metrics are reported and real-time deployment is not addressed despite near-operational use.	2 Anomaly masks are spatially explicit and shown alongside chlorophyll-a maps, making outputs visually interpretable. However, internal variance parameters offer limited insight, and species identification always requires follow-up in situ analysis.	3 It directly addresses harmful algal bloom detection and eutrophication monitoring, with significant implications for fisheries and marine biodiversity, validated against ecological observations.
Wu et al. [41]	2 The full red tide framework reports strong performance (average OA 97.66%), with cross satellite tests, sensitivity analysis, and comparison with seven methods. However, the Isolation Forest AD stage is not evaluated separately, so validation is supportive but not isolated.	2 The Isolation Forest stage uses an established one-class method within a clear three stage verification pipeline. Training sample formation and threshold setting are explained, but there is limited theoretical justification for this IF configuration.	2 Isolation Forest is computationally practical, and applying it only to candidate pixels reduces computational burden. The overall method has acceptable runtime, although no runtime or scalability analysis specific to IF is reported.	2 The AD stage provides anomaly scores, a defined threshold, and clear spatial detection maps. The wider framework is physically interpretable, but internal Isolation Forest decisions are not explained, giving moderate interpretability.	3 The AD stage directly identifies <i>Noctiluca scintillans</i> red tide regions in real satellite imagery across several platforms, supporting harmful algal bloom monitoring, marine ecological protection, and disaster response.
Chuang et al. [42]	2 Precision, interval recall (IR), and F1-IR are explicitly reported across multiple anomaly thresholds, with Monte Carlo baselines and Bonferroni-corrected significance tests. The best early-warning result is modest (precision 22.9%, IR 40.7%, F1-IR 0.293), and the authors acknowledge limited observations and only six ponds, so validation is useful but not extensive.	2 The anomaly detection design is grounded in established time-series theory: stationarity is tested with ADF, stationary sequences use a bootstrap-based R_{mean} threshold, and non-stationary sequences use an LSTM residual approach. The methods are clearly justified from prior work, but the paper mainly adapts established techniques rather than developing a new theoretical framework.	2 The LSTM is small (one layer with seven hidden units) and operates on daily pond records, while the stationary-sequence method uses straightforward sliding windows and bootstrap thresholds. No runtime, memory, or scalability benchmarks are reported, but the implementation appears computationally manageable for this dataset and similar aquaculture monitoring tasks.	2 Detected anomalies are shown against water-colour changes, and the residual/threshold logic makes the anomaly signal understandable. The authors also interpret the timing of water-temperature and pH anomalies ecologically. However, the internal LSTM decision process remains largely a black box, so interpretability is moderate rather than fully transparent.	3 The anomaly detection is directly tied to real grouper-pond monitoring and to early warning of undesirable water-colour changes. The study links detected water-quality anomalies to practical pond management and autonomous aquaculture monitoring, giving the method clear ecological and operational relevance.
Hammond et al. [43]	2 Extreme upwelling events are identified from historical grid cell distributions using quantile thresholds after K-means delineation of upwelling regions. The method is tested on held-out and near real time 2021 data but lacks in situ ground truth and an identified optimum threshold, so validation is reasonable but not conclusive.	3 The implementation is strongly grounded in oceanographic theory. Upwelling is represented by high chlorophyll-a and low SST, clustering is assessed with the elbow criterion and Calinski-Harabasz score, and thresholds are derived from long-term local historical distributions.	2 K-means and quantile thresholding are lightweight, and the model is fitted on only 15% of the historical data before being applied to the remainder and near real time observations. This supports operational use, although no runtime or scalability benchmarks are reported.	3 The anomaly logic is highly transparent, with the upwelling core, surrounding regions, and extremes shown on spatial maps and thresholds tied directly to historical quantiles. The authors also explain where false identification may arise from eddies or fronts.	3 The method addresses ecologically important coastal upwelling variability in the Western Indian Ocean and supports an operational Upwelling Watch service, with outputs linked to fisheries, food security, ecosystem health, and management decisions.